

535/1
PHYSICS
Paper 1
(theory)
Marc/Apr. 2026
1 ¼ hours



**UGANDA MARTYRS HIGH SCHOOL
KIBOGA**

S.1 Assessment II examinations

PHYSICS

Paper 1

(theory)

1 hours 15 minutes

Instructions to the candidates

*This paper consists of **three** examination items*

*You are required to select **two (2)** items in all*

*Any additional **item(s)** responded shall not be scored*

In your presentations, make clear drawings, neat hand writing and clear substitutions

Non programmable calculators and tables may be used.

At the end of the paper, fasten all your work together

Turn over

Item 1

The S.1 learners went to the physics laboratory for a lesson. The teacher had prepared the laboratory apparatus and materials for the learners to carry out an experiment in groups. However, the learners entered while running trying to get stools and ended up colliding with one another and breaking some of the apparatus.

In addition, the apparatus prepared were materials like a small irregular stone, water and a small regular wooden block of **mass 800g** and dimensions **15.0cm x 10.0cm x 6.0cm**.

Hint (the density of water = 1gcm^{-3})

Task

As a learner of physics;

- (a) Guide the S.1 learners on how they should conduct themselves on entering the laboratory and while in the laboratory.
- (b) Describe how you would carry out the scientific investigation to determine the density of the stone.
- (c) (i) Calculate the density of the wooden block provided in kgm^{-3} and express your answer in scientific notation.
(ii) Explain what would be observed if the wooden block in (c)(i) above was immersed in water.

Item 2

Your friend needs to take **10 books** to class each day for the lessons. His parents had bought for him **20 black books** each of dimensions **30cm by 21cm by 2cm** and he was given a bag of **volume 10080cm³**. He was really wondering whether the required books would fit in his bag. He reported late when you had already studied about fundamental and derived quantities so he needed a clear explanation about them.

Task

As a learner of physics;

- (a) Differentiate between the two quantities and give two examples in each case.
- (b) Determine if the books will fit in the bag and how many books will remain outside the bag.
- (c) If your friend sits in the single sitter of upper deck of dimensions **105cm by 18cm**, how many books can he put on the upper deck at a time.

Item 3

Ten identical golden coloured cubes were put in the market for sale for **Ugx40, 000** for each gramme incase they were gold nuggets. As they tried to prove that they were gold, the mass of one cube was found to be **0.1568kg** and the side of the cube was **2cm** in length each.

Hint: The density of gold is **19.3gcm³**.

Task

As a learner of physics;

- (a) Determine if the cubes are made of gold.
- (b) Determine the amount of money that would be got if all the 10 cubes were sold.

END

“The Lord is risen indeed, halleluiah”